

1. Random

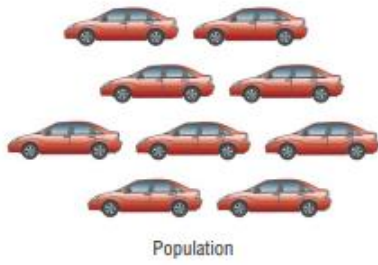
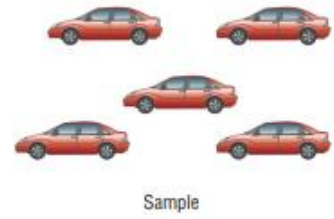
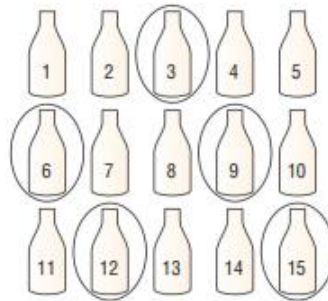


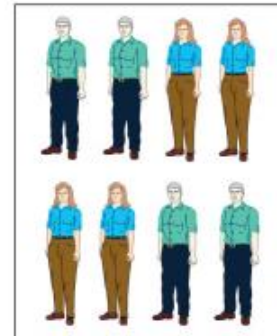
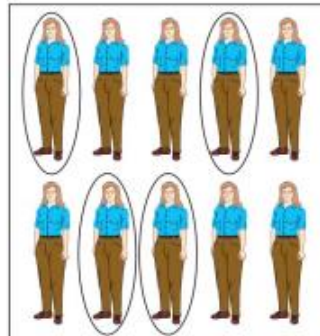
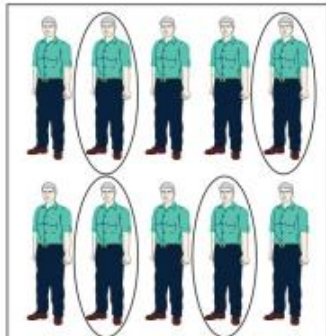
Table D	Random Numbers			
10480	15011	01536	02011	81647
22368	46573	25595	85393	30995
24130	48360	22527	97265	76393
42167	93093	06243	61680	07856
37570	39975	81837	16656	06121
77921	06907	11008	42751	27750
99562	72905	56420	69994	98872
96301	91977	05463	07972	18876
89579	14342	63661	10281	17453
85475	36857	43342	53988	
28918	69578	88321		
63553	40961			



2. Systematic



3. Stratified



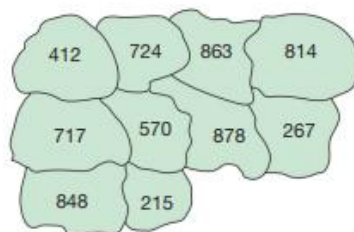
Men

Women

Sample

Population

4. Cluster



Population



Sample

TABLE 1–4 Summary of Sampling Methods

Random	Subjects are selected by random numbers.
Systematic	Subjects are selected by using every k th number after the first subject is randomly selected from 1 through k .
Stratified	Subjects are selected by dividing up the population into subgroups (strata), and subjects are randomly selected within subgroups.
Cluster	Subjects are selected by using an intact subgroup that is representative of the population.

EXAMPLE 1–5 Sampling Methods

State which sampling method was used.

- Out of 10 hospitals in a municipality, a researcher selects one and collects records for a 24-hour period on the types of emergencies that were treated there.
- A researcher divides a group of students according to gender, major field, and low, average, and high grade point average. Then she randomly selects six students from each group to answer questions in a survey.
- The subscribers to a magazine are numbered. Then a sample of these people is selected using random numbers.
- Every 10th bottle of Super-Duper Cola is selected, and the amount of liquid in the bottle is measured. The purpose is to see if the machines that fill the bottles are working properly.

SOLUTION

- Cluster
- Stratified
- Random
- Systematic

Experimental Design

There are several different ways to classify statistical studies. This section explains two types of studies: observational studies and experimental studies.

In an **observational study**, the researcher merely observes what is happening or what has happened in the past and tries to draw conclusions based on these observations.

Example of an Observational Study:

Imagine researchers want to understand how smartphone ownership has evolved over time among adults. They conduct the following:

Data Collection:

In 2005, they survey a large, representative sample of adults about whether they own a smartphone. They repeat a similar survey in 2020.

Comparing Time Points:

The collected data reveal that in 2005, only about 10% of adults reported owning a smartphone, whereas by 2020, that number had increased to 80%. Furthermore, they notice that while younger adults had high ownership rates early on, there was a notable increase among older adults in 2020.

In an **experimental study**, the researcher manipulates one of the variables and tries to determine how the manipulation influences other variables.

Example of an experimental Study:

Let's consider a researcher group want to find the impact of goal specificity on performance. The process is:

Groups and Intervention:

The researchers divided the participants (female undergraduate students) into two groups. One group (the control group) received vague instructions—"Do your best"—while the other group (the experimental group) was given a specific goal—to increase the number of sit-ups by 10% each day.

Independent Variable:

The independent variable is the type of instruction provided:

Vague instruction ("Do your best")

Specific goal (increase sit-ups by 10% each day)

Dependent Variable:

The dependent variable is the performance measured as the number of sit-ups completed in 90 seconds.

Outcome:

After four days, the group with the specific goal averaged 56 sit-ups, while the group with the vague instruction averaged 43 sit-ups. This significant difference in performance indicates that having specific, measurable goals led to better outcomes.

Conclusion:

The study concludes that athletes (or in this case, students) perform better when given specific, quantifiable targets compared to when they are simply told to "do your best." Since the only difference between the two groups was the type of instruction (the manipulated variable), the study provides evidence that the specific goal-setting was the factor leading to improved performance.

The **independent variable** in an experimental study is the one that is being manipulated by the researcher. The independent variable is also called the **explanatory variable**. The resultant variable is called the **dependent variable** or the **outcome variable**.

EXAMPLE 1-6 Experimental Design

Researchers randomly assigned 10 people to each of three different groups. Group 1 was instructed to write an essay about the hassles in their lives. Group 2 was instructed to write an essay about circumstances that made them feel thankful. Group 3 was asked to write an essay about events that they felt neutral about. After the exercise, they were given a questionnaire on their outlook on life. The researchers found that those who wrote about circumstances that made them feel thankful had a more optimistic outlook on life. The conclusion is that focusing on the positive makes you more optimistic about life in general. Based on this study, answer the following questions.

- a.* Was this an observational or experimental study?
- b.* What is the independent variable?
- c.* What is the dependent variable?
- d.* What may be a confounding variable in this study?
- e.* What can you say about the sample size?
- f.* Do you agree with the conclusion? Explain your answer.

SOLUTION

- a.* This is an experimental study since the variables (types of essays written) were manipulated.
- b.* The independent variable was the type of essay the participants wrote.
- c.* The dependent variable was the score on the life outlook questionnaire.
- d.* Other factors, such as age, upbringing, and income, can affect the results; however, the random assignment of subjects is helpful in eliminating these factors. See the answer for the next question.
- e.* In this study, the sample uses 30 participants total.
- f.* Answers will vary.

Exercises 1–4

For Exercises 15–17, determine whether an observational study or an experimental study was used.

15. Subjects are randomly assigned to three groups.
Group 1 receives a placebo daily, group 2 receives a high dose of vitamin B daily, and group 3 receives a multivitamin daily. After 3 months, each participant is

asked how many incidents of depression that she or he had, and the occurrences were compared.
16. In a large city, a researcher decides to check the police department records for 2 months to determine what types of vehicles were stolen.
17. In a public park, visitors were asked how many minutes a day they walked their dogs.

In Exercises 19–21, identify the independent variable and the dependent variable.

19. According to the *British Journal of Sports Medicine*, a regular 30-minute workout could slash your risk of catching a cold by 43%.
20. The *Journal of Behavioral Medicine* reported that sharing a hug and holding hands can limit the physical effects of stress such as soaring heart rate and elevated blood pressure.
21. A research study stated that meditation helps people make more-rational decisions.